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Mounting apparatus for dynamically loaded structural joints

Abstract

A mounting structure apparatus for dynamically loaded structure joints is disclosed. The apparatus includes a first member having an aperture through a first surface and communicating with a counterbore in a second surface. A second member is included having a blind bore coaxial with the first member aperture. An insert is engageable with the counter bore, and a fastener is disclosed insertable into the first member aperture and interactive with the second member to join the first member to the second member.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation-in-part of U.S. patent application Ser. No. 15/950,582, filed Apr. 11, 2018, and the entire contents and disclosures of which are incorporated herein by reference.

TECHNICAL FIELD

(1) This disclosure relates to improved mounting structures for dynamically loaded structural joints. The mounting structure allows shear strength and tensile strength of a joint to be independently tuned by varying sizes of an engaged counter bore and a fastener.

BACKGROUND

(2) Existing fasteners are limited in creating a joint with improved shear and tensile strength. Traditional joints consisting of a bolt threaded into a nut are limited in the amount of energy they can absorb prior to failure. This is especially important in situations with dynamically loaded structural joints. Dynamically loaded structural joints, such as those formed when jointing armor plating, are subjected to shear and tensile forces. Existing fasteners can also result in a “secondary projectile” when subjected to shear forces that overcome the fastener strength. It is important to be able to accommodate these forces in armored vehicles which may be subjected to blast forces that may overcome existing fasteners.

(3) Improvements to such joints are given in the several embodiments disclosed. In one embodiment, one piece of a structure may be attached to another piece of a structure with a bolt and an insert that is mechanically loaded as a result of translational movement of the two pieces. The insert may be threaded into the main piece of structure, and the insert may have internal threads for attachment, and a shoulder feature that sits on the surface of the main structure. The mating piece of the structure has a counterbore that fits over the shoulder of the insert and transfers load when translation occurs between the two structures.

SUMMARY

(4) According to an exemplary aspect of the present disclosure, a mounting structure apparatus includes a first member having a first surface and a second surface opposite to the first surface, and having an aperture formed through the first surface and a counterbore formed with a circumferential wall in the second surface, the counterbore communicating with the aperture, a second member having a bore facing the second surface of the first member and being coaxial with the aperture of the first member, an insert having a shoulder and also being engageable with the counterbore of the first member and the bore of the second member, and a fastener insertable into the aperture of the first member and coupled with the bore of the second member by engaging the insert to join the first member to the second member. Further, the circumferential wall of the counterbore is formed with a closed shape and configured to move into a direct engagement with a radial end surface of the shoulder in the insert.

(5) According to a further aspect of the present disclosure, the insert forms a threaded bore and external threads, and the external insert threads are engageable with threads in the bore of the second member. Further, the threaded bore engages the fastener. The counterbore of the first member is formed with an oblong shape in a direction defined as X-axis. The aperture of the first

member is formed with an elongated slot along the X-axis and a slot length of the aperture is smaller than a length of the counterbore in the X-axis.

(6) According to a further aspect of the present disclosure, the counterbore has a first dimension to accommodate the insert and a second dimension smaller than the first dimension and the second dimension is separated from the first dimension by a length L in a direction defined as X-axis such that the counterbore of the first member is tapered from the first dimension to the second dimension in the X-axis. The second dimension is smaller than a diameter of the insert whereby the insert interferes with the counterbore in the event that the first and second members are relatively moved from each other.

(7) According to another aspect of the present disclosure, a mounting structure apparatus includes a first member having a first surface and a second surface opposite to the first surface, the first member having an aperture being formed through the first surface and communicating with a counterbore of the second surface, a second member having a bore facing the second surface of the first member and being coaxial with the aperture of the first member, an insert having a blind bore and being engageable with the counterbore of the first member and the bore of the second member, and a fastener insertable into the aperture of the first member and coupled with the bore of the second member by engaging the insert to join the first member to the second member. Further, the counterbore of the first member is formed with a tapered slot having a first dimension to accommodate the insert and a second dimension smaller than the first dimension, and the second dimension is separated from the first dimension by a length.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is a cross-sectional view of one embodiment of the mounting apparatus;
- (2) FIG. 2 is a cross-sectional view of another embodiment of the mounting apparatus;
- (3) FIG. 3 is a cross-sectional view of another embodiment of the mounting apparatus;
- (4) FIG. 4 is a cross-sectional view of another embodiment of the mounting apparatus;
- (5) FIG. 5 is a cross-sectional top view of the mounting apparatus showing an insert within a counterbore in the apparatus of FIG. 3;
- (6) FIG. 6 is a cross-sectional side view of another embodiment of the mounting apparatus with a counterbore;
- (7) FIG. 7 is a perspective top view of a plate having an aperture and a counterbore of FIGS. 3 and 6; and
- (8) FIG. 8 is a perspective bottom view of a plate having an aperture and a counterbore of FIGS. 3 and 6.

DETAILED DESCRIPTION

- (9) All figures and examples herein are intended to be non-limiting; they are mere exemplary iterations and/or embodiments of the claims appended to the end of this description. Modifications to structure, materials, the order of steps in procedures, etc., are contemplated.
- (10) Referring now to the drawings, and particularly to FIG. 1, there is shown a cutaway side view of one embodiment of the mounting apparatus. Specifically, mounting structure **10** is shown joining first plate **12** and second plate **14**. Plate **14** is equipped with an aperture **16** extending there through, and oriented to be coaxial and overlies blind bore **18** in plate **12**. Member or plate **12** has insert projection **20** which may be threaded with an internal thread **22** in the bore. Plate **14** has a circular counter bore **24**, so that the plate **14** may be moved into engagement with the insert projection **20** so as to engage shoulder **26** on plate **12** to shoulder **28** on plate **14**. The counterbore is designed to have a depth D to accommodate at least a portion of the projection **20**. A fastener **32**, which may be threaded **34**, and a washer **30**, are used to secure the plates together. Specifically, in this

embodiment, plate **14** is placed adjacent to plate **12**, and projection **20** extends into the counterbore **24**, and the plate **14** is moved toward plate **12** until shoulder **28** on plate **14** engages shoulder **26** on plate **12**. Threaded fastener **32** is passed through washer **30** and is inserted through aperture **16** until it threadably engages threads **22** in the blind bore **18**. The fastener is threadably inserted into threaded engagement until the head **33** of the fastener is in engagement with the washer, thereby securing the plate in place. The mounting structure as described imparts improved structural integrity to dynamically loaded joints.

(11) FIG. **2** is a different embodiment of the mounting apparatus. Specifically, mounting structure **36** is shown joining two plates **38** and **40**. Plate **38** is equipped with a blind bore **42** and plate **40** is equipped with a circular counterbore **44**. An insert **46** having external threads **45** is threaded into engagement with the threaded bore **42** in plate **38** until insert shoulder **48** engages the surface **39** of plate **38**. The insert shoulder is of such dimension as to be accommodated within the counterbore in plate **40**. Plate **40** is placed in adjacent position to plate **38** such that surface **41** is adjacent to surface **39**. A threaded fastener **54**, having a head **55**, is passed through a washer **52**, passed through aperture **51** in plate **40**, and threadably engaged with internal threads **47** in the insert, thereby securing the two plates to form a dynamically loaded structural joint.

(12) FIG. **3** is another embodiment of the mounting apparatus. In this embodiment, mounting apparatus **56** comprises two plates **58** and **60**, held in clamping relationship by fastener **61**. Fastener **61**, which may have threads **57**, may be passed through washer **65**. Plate **58** has an aperture **66** through which the fastener **61** may be inserted. Plate **60** is equipped with a threaded blind bore **62**, into which may be threaded insert **64**. The insert **64** has a threaded aperture **63** into which the fastener **61** may be cooperatively threaded. Insert **64** has shoulder **67** which, when the insert is fully threaded into the bore, surface **70** on the shoulder may be (but not necessarily) brought into engagement with surface **72** of plate **60**. The shoulder **67** is dimensioned such that together they form a diameter or dimension **74**, which is accommodated within offset counterbore **76** in plate **58**. While FIG. **5** describes the counterbore in plate **58**, it is understood that such a counterbore may be in both plates. As seen in FIG. **5**, counterbore **76** has a first end **78** with a first diameter or dimension **80** separated by a length **L** from a second end **82**, with a second diameter or dimension **84**. Diameter or dimension **80** is greater than size than diameter or dimension **84**. Diameter or dimension **80** is of a sufficient size to accommodate the diameter or dimension **74** of threaded insert **64**. In this regard, the counterbore may be oblong shape along X-axis (i.e., a direction **90** in FIG. **5**). In other embodiments, it is not necessary to have first and second diameters or dimensions in the counterbore **76**. In some embodiments, it is possible to have a counterbore that has angular dimensions, such that the first end is of sufficient size to accommodate the diameter or dimension **74**, and is of a larger dimension than the second end. When the apparatus of FIGS. **3** and **5** is subjected to shear forces **88**, plate **58** moves in the direction **90** (i.e., along the X-axis). As plate **58** moves in the direction **90** (along the X-axis), the offset counterbore moves into engagement with the diameter or dimension **74** of the threaded insert **64**. As the force moves the counterbore wall **125** into engagement with the insert diameter or dimension, shear force is dissipated and is energy is absorbed. Material from plate **58** may be deformed in the process to absorb the interaction of the threaded diameter or dimension and the counterbore wall **125** without causing the mounting apparatus to come apart, thereby providing for a dynamically mounted joint. The embodiment of FIG. **3** may result in different clamping arrangements between the plates and the insert and the fastener. One possibility is that, with the insert fully threaded into the bore, and fastener plate **58** clamps with plate **60**, and surfaces **70** and **72** are engaged with each other, but surface does not clamp with insert **68**. In another embodiment, plate **58** clamps with plate **60** and surfaces **70** and **72** are in engagement with each other. In yet another embodiment, plate **58** and **60** are clamped together, and the threaded insert may engage the counterbore surface **59**.

(13) FIG. **4** is another embodiment of the mounting structure of FIGS. **3** and **5**, wherein plate **92** has a counter bore **94** to accommodate the insert **96**. In this embodiment, the insert, which may be

threaded, engages both the counterbore **94** of plate **92**, but also engages counterbore **98** of plate **100** when the apparatus is subjected to dynamic forces as described with reference to FIG. **3** and FIG. **5**. Thus the insert as shown in FIG. **4** does not experience stress riser issues at the insert shoulder, and when assembled, the mounting apparatus couples plates **100** to **92** through the insert **96** when the fastener **49** is passed through aperture **17** and threaded into the insert as at **19**. It is understood that in this embodiment, both plate **92** and **100** may have a counterbore similar to that as described in FIG. **5**. Those skilled in the art also appreciate that the tapered slot as described in the previous figures may both be rounded counterbores in plates **92** and **100**.

(14) FIG. **6** is a cross-sectional view of a dynamically loaded joint, showing a side view of the interaction of the threaded insert diameter and a pocket (which has similar configurations of FIGS. **3** and **5**). For example, in this embodiment of the present disclosure, the mounting apparatus comprises two plates (i.e., two members) **102** and **124**, held in clamping relationship by fastener **61** as shown in FIG. **3**. The fastener **61** having threads may be passed through washer **65**. The plate **124** has an aperture **126** through which the fastener **61** may be inserted. Further, plate **124** has an elongated slot **126** and a counterbore (see FIGS. **7** and **8**) to receive the fastener **61** and the insert **106** respectively, and plate **102** has an aperture **104** into which is placed insert **106**. In accordance with another embodiment of the present disclosure, the plate **102** may have a blind bore (for example, see FIG. **3**). Insert **106** has shoulders **108** and **110**, with surfaces **112** and **114** which, when the insert is placed into the aperture **104**, engage surface **116** on plate **102**. Those skilled in the art understand that both shoulders **108** and **110** are formed by the diameter **115** of the insert **106**, and are called out as separate numbers merely for clarity. Similarly, the surfaces depicted as **112** and **114** are actually the same surface in a top view, but are called by separate reference numbers in this side view.

(15) The shoulders form an insert diameter or dimension **115**, which is accommodated by a first diameter or dimension **117** in a pocket **118** (i.e., a counterbore). The pocket **118** has a second diameter or dimension **120** separated from the first diameter or dimension by a length **122**. The second diameter or dimension is smaller than the first diameter or dimension. In this regard, the structure as described is also seen in FIG. **5**. Motion between the two plates **102** and **124** causes the diameter or dimension of the insert to contact and interact with the side walls **125** (see FIG. **5**) in the pocket until the motion of one plate relative to the other is stopped by interaction of the insert with the sides of the pocket. Further, the aperture **126** of the plate **124** is formed with an elongated slot along the direction **90** (i.e., X-axis) such that an impact force causes the first plate to move relative to the second plate. Accordingly, the insert **106** moves within the tapered counterbore from the first dimension **117** to the second dimension **120** along the direction **90** (i.e., X-axis), and the fastener also moves within the elongated slot along the X-axis. In general, the slot length of the aperture **126** is smaller than the length of the pocket **118** (counterbore) in the X-axis. As the one plate moves in the direction **90** (i.e., along the X-axis) in relation to the other, it encounters shear forces **88** (as shown in FIG. **5**) until the energy is absorbed, thereby relieving the motion between the plates without separating the dynamically loaded structure.

(16) Any of the described mounting structures may be used for joining dynamically loaded structures. One piece of structure is attached to another piece of structure with a bolt and an insert that is mechanically loaded as a result of translational movement of the two pieces of structure. The insert may be threaded or otherwise joined, such as a physical/chemical bonding, or a press fit, or a friction weld) into the main piece of structure, and, may have internal threads for attachment and a shoulder feature that sits on the surface of the one of the structures. The other piece of the structure (mating structure) has a counterbore that fits over the shoulder of the insert that transfers load when translation occurs along the X-axis (i.e., a transverse direction relative to the insertion direction (Z-axis) of the fastener between the two structures (members) to be joined. A fastener, such as a bolt, threads into the internal thread of the insert holding two pieces of structure together along the Z-axis. In one embodiment, the configuration of the counterbore depth is tolerance such that the

clamp load provided by the bolt clamps the two members together. The counterbore may be oblong, tapered or circular shape along the X-axis. An oblong counterbore allows for the members to be joined at an angle and will load the fastener and counterbore in three directions and the fastener in one direction. The mounting structures as described may be tuned as one might tune a shear pin, so that it could be constructed to take certain forces without failure but fail if those forces are exceeded.

(17) The tolerances of the fastener, insert major diameter and counterbore diameter can be utilized to adjust the load sharing between the fastener (bolt) and the insert. The fastener can be utilized to adjust the load sharing between the fastener and the insert. The insert can be comprised of different materials dependent on the properties required. To maximize the energy the joint can transmit, a fastener with superb elongation and energy to failure properties may be selected.

(18) The mounting structures as described may be suited for structural joints and may be selected for use in the application of under armor to eliminate the need for a puck on the inside face of the structure. The insert is threaded into the structure and can be replaced without reworking the base structure. The mounting apparatus as described can be utilized to reduce weight by allowing for the use of smaller fasteners (bolts). In addition the shear strength and tensile strength of the joint can be tuned independently by varying the sizes of the engaged counterbore and the fastener (bolt). Joints may be designed to have different shear strength in different directions by shaping of the counterbore.

(19) Although the steps of the above-described processes have been exemplified as occurring in a certain sequence, such processes could be practiced with the steps performed in a different order. It should also be understood that certain steps could be performed simultaneously, that other steps could be added, or that certain steps could be omitted. In other words, the descriptions of the processes are provided for the purpose of illustration, and should not limit the claimed invention.

(20) Accordingly, it is to be understood that the above description is intended to be illustrative and not restrictive. Many embodiments and applications other than the examples provided would be apparent upon reading the disclosure. The scope of the invention should be determined with reference to the appended claims along with the full scope of equivalents. It is intended that future developments will occur, and that embodiments of the disclosed systems and methods will incorporate and be incorporated with such future developments.

(21) Use of singular articles such as “a,” “the,” “said” together with an element means one or more of the element unless a claim expressly recites to the contrary.

Claims

1. A mounting structure apparatus, comprising: a first member having a first surface and a second surface opposite to the first surface, the first member having an elongated slot formed in the first surface along a longitudinal axis (X-axis) and a counterbore formed in the second surface along the longitudinal axis (X-axis) such that the X-axis of the elongated slot is aligned with the X-axis of the counterbore; a second member having a bore facing the second surface of the first member and communicating with the counterbore of the first member; an insert having a shoulder and a blind bore, the insert engageable with the counterbore of the first member and the bore of the second member; and a fastener configured to insert into the elongated slot of the first member and engaged with the blind bore of the insert such that the fastener is indirectly coupled with the bore of the second member to join the first member to the second member, wherein the counterbore of the first member is formed with a peripheral wall enclosed by the second surface, and the shoulder of the insert and the fastener are configured to slide along the longitudinal axes of the elongated slot and the counterbore of the first member, and wherein the counterbore has a first dimension to accommodate the insert and a second dimension smaller than a diameter of the insert, and the second dimension is separated from the first dimension by a length L along the X-axis such that the

- counterbore of the first member is tapered from the first dimension to the second dimension along the X-axis, and whereby the insert is configured to interfere with the counterbore in the event that the first and second members are moved relative to each other.
2. The mounting structure apparatus of claim 1, wherein the insert forms a threaded bore and external threads, the external insert threads engageable with threads in the bore of the second member, and the threaded bore engages the fastener.
 3. The mounting structure apparatus of claim 1, wherein the counterbore of the first member is formed with an oblong or elongated tapered shape along the X-axis.
 4. The mounting structure apparatus of claim 1, wherein a slot length of the aperture elongated slot is smaller than a length of the counterbore along the X-axis.
 5. The mounting structure apparatus of claim 1, wherein the elongated slot of the first member comprises a peripheral wall fully enclosed by the first surface.
 6. A mounting structure apparatus, comprising: a first member having a first surface and a second surface opposite to the first surface, the first member having an elongated slot formed in the first surface along a longitudinal axis and aligned with a counterbore formed in the second surface along the longitudinal axis; a second member having a bore facing the second surface of the first member and communicating with the counterbore of the first member; an insert having a blind bore, the insert engageable with the counterbore of the first member and the bore of the second member; and a fastener configured to insert into the elongated slot of the first member and engaged with the blind bore of the insert such that the fastener is indirectly coupled with the bore of the second member to join the first member to the second member, wherein the counterbore of the first member is formed with a peripheral wall enclosed by the second surface, and the insert and the fastener are configured to slide along the longitudinal axes of the elongated slot and the counterbore of the first member, and wherein the counterbore of the first member is formed with an elongated tapered slot having a first dimension to accommodate the insert and a second dimension smaller than a diameter of the insert, and the second dimension is separated from the first dimension by a length along the longitudinal axis, and whereby the insert is configured to interfere with the counterbore in the event that the first and second member are moved relative to each other.
 7. The mounting structure apparatus of claim 6, wherein a slot length of the elongated slot is smaller than a length of the counterbore in along the X-axis.
 8. The mounting structure apparatus of claim 6, wherein the elongated slot of the first member comprises a peripheral wall fully enclosed by the first surface.
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